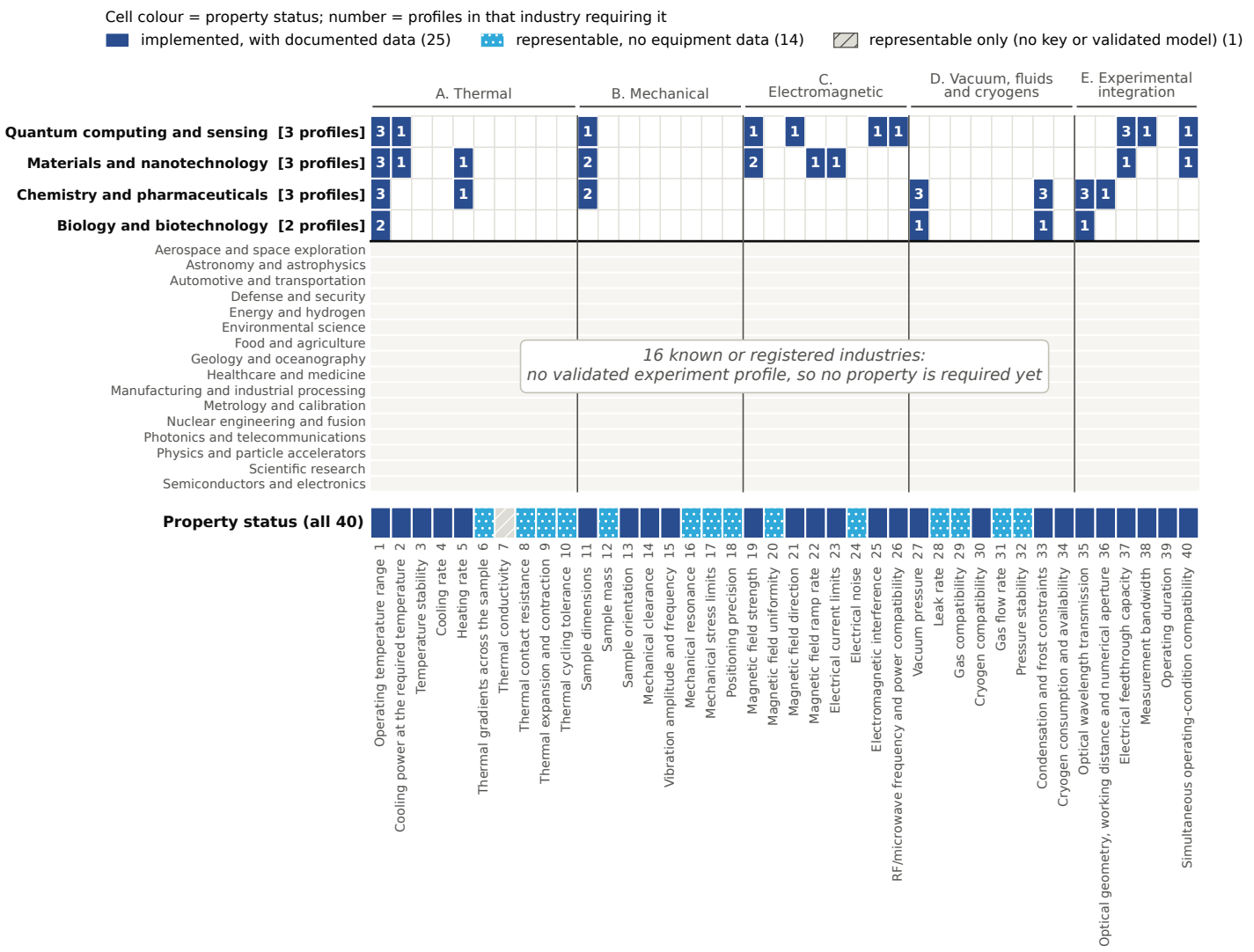

Cell colour = property status; number = profiles in that industry requiring it
implemented, with documented data (25)
representable, no equipment data (14)
representable only (no key or validated model) (1)
A. Thermal
B. Mechanical
C. Electromagnetic
D. Vacuum, fluids and cryogens
E. Experimental integration
Quantum computing and sensing [3 profiles]
Materials and nanotechnology [3 profiles]
Chemistry and pharmaceuticals [3 profiles]
Biology and biotechnology [2 profiles]
Aerospace and space exploration
Astronomy and astrophysics
Automotive and transportation
Defense and security
Energy and hydrogen
Environmental science
Food and agriculture
Geology and oceanography
Healthcare and medicine
Manufacturing and industrial processing
Metrology and calibration
Nuclear engineering and fusion
Photonics and telecommunications
Physics and particle accelerators
Scientific research
Semiconductors and electronics
16 known or registered industries:
no validated experiment profile, so no property is required yet
Property status (all 40)
1 Operating temperature range
2 Cooling power at the required temperature
3 Temperature stability
4 Cooling rate
5 Heating rate
6 Thermal gradients across the sample
7 Thermal conductivity
8 Thermal contact resistance
9 Thermal expansion and contraction
10 Thermal cycling tolerance
11 Sample dimensions
12 Sample mass
13 Sample orientation
14 Mechanical clearance
15 Vibration amplitude and frequency
16 Mechanical resonance
17 Mechanical stress limits
18 Positioning precision
19 Magnetic field strength
20 Magnetic field uniformity
21 Magnetic field direction
22 Magnetic field ramp rate
23 Electrical current limits
24 Electrical noise
25 Electromagnetic interference
26 RF/microwave frequency and power compatibility
27 Vacuum pressure
28 Leak rate
29 Gas compatibility
30 Cryogen compatibility
31 Gas flow rate
32 Pressure stability
33 Condensation and frost constraints
34 Cryogen consumption and availability
35 Optical wavelength transmission
36 Optical geometry, working distance and numerical aperture
37 Electrical feedthrough capacity
38 Measurement bandwidth
39 Operating duration
40 Simultaneous operating-condition compatibility